

## ***Dr Ahmed Magdi Moustafa***

**Co-Director Center for Microbial Medicine at the Children's Hospital of Philadelphia and  
Assistant Professor of Pediatrics at The University of Pennsylvania**

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### **PERSONAL INFORMATION**

Born: 09/16/1987, Alexandria, Egypt

Citizenship: Egyptian (US Permanent Resident)

### **EDUCATION**

**Doctor of Philosophy Biomedical Sciences** (Jan 2011-Aug 2015)

School of Veterinary and Life Sciences, Murdoch University, Perth, Australia.

- Thesis: A Molecular Epidemiological Study of Haemorrhagic Septicaemia Disease in Pakistan.
- Supervisors: Prof. John Edwards, Dr. Mark Bennett, Dr. Tim Hyndman and Prof. Stan Fenwick.

**Bachelor of Pharmaceutical Sciences** (2004-2009) - Grade High Distinction

Faculty of Pharmacy, Alexandria University, Alexandria, Egypt.

- Graduated as the **ninth highest ranked** student in a class of 1500 students.
- Appointed as a tenure-track faculty upon graduation.

### **RESEARCH EXPERIENCE**

**Founder and Co-Director Center of Microbial Medicine at CHOP (July 2024 -Present)**

Children's Hospital of Philadelphia, Philadelphia, PA, USA.

**Assistant Professor, Department of Pediatrics (July 2022-Present)**

Perelman School of Medicine, University of Pennsylvania and Children's Hospital of Philadelphia, Philadelphia, PA, USA.

**Director Microbiome Sequencing Core (Sept 2021-Present)**

CHOP Microbiome Center, Division of Gastroenterology, Hepatology, and Nutrition, Children's Hospital of Philadelphia, Philadelphia, PA, USA.

**Co-Director microbial ARchive and Cryo-collection ([microbialARC](#)) (Mar 2021-Present)**

Division of Gastroenterology, Hepatology, and Nutrition, Children's Hospital of Philadelphia, Philadelphia, PA, USA.

**Scientist I (PennCHOP Microbiome Program) (Mar-Sept 2021)**

Division of Gastroenterology, Hepatology, and Nutrition, Children's Hospital of Philadelphia, Philadelphia, PA, USA.

**Visiting scientist (Mar 2017-Ongoing)**

Sackler Institute for Comparative Genomics, American Museum of Natural History, NY, USA.

**Post-doctoral Fellow (Infectious Diseases) (Sep 2016-Mar 2021)**

Paul Planet Lab, Children's Hospital of Philadelphia, Philadelphia, PA, USA.

**Research Assistant (Oct 2015-January 2017)**

School of Veterinary and Life Sciences, Murdoch University, Perth, Australia.

**Research Collaborator (Aug-Oct 2013)**

Department of Microbiology, Monash University, Melbourne, Australia.

**Doctoral Researcher (Jun 2011-Oct 2014)**

School of Veterinary and Life Sciences, Murdoch University, Perth, Australia.

## **SECONDARY AFFILIATIONS**

**Department of Biomedical and Health Informatics, Children's Hospital of Philadelphia, Philadelphia, PA, USA (September 2022-Present)**

**Department of Microbiology, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA (July 2023-Present)**

## **TEACHING AND MANAGING EXPERIENCES**

**PhD Mentor (Jul 2022-Ongoing)**

Children's Hospital of Philadelphia, University of Pennsylvania, Philadelphia, PA, USA.

- **Co-mentoring Qianxuan She (MVP student) with Paul Planet and Joe Zackular.**

**Thesis Committees (Jul 2022-Ongoing)**

Children's Hospital of Philadelphia, University of Pennsylvania, Philadelphia, PA, USA.

- **Stephanie Higgins: Chair of Thesis Committee and Mentor for Master of Science in Microbiology and Immunology, Jefferson University.**
- **Jiayi Duan: Thesis Committee Member (CAMB MVP)**
- **Natalie Hagen: Chair of Thesis Committee (CAMB MVP)**

**Co-mentoring and supervising (Sep 2016-Ongoing)**

Children's Hospital of Philadelphia, University of Pennsylvania, Philadelphia, PA, USA.

- Actively co-mentoring and supervising undergraduate and graduate students, research technicians, postdocs and visiting researchers.
- Co-Mentor and Supervisor for trainees, Children's Hospital of Philadelphia (CRISP, RISES) and UPenn (PURM, CURF).
  - UPenn/CHOP Mentees: Arnav Lal, Lidiya Denu, Kelsey O'Brien, Prioty Ferheen Sarwar, Andries Feder, Daniel Morreale, Alena Nag, Elizabeth Qian, Lauren Hamlette, Carolyne Everett, Anas Soliman, Sobhika Sampath, Srujan Yamali, Noor Qureshi, Shivani Shah.
  - Visiting Mentees: Martinez Rodrigo (Chile), Stephanie Rodrigues (Brazil), Nicolas Aguilera (Chile), Qian Zhou (Denmark).
- Supervising the researchers through their first hands on experience in genomics and bioinformatics and walk them through different pipelines.

**Microbial Genomics Journey Workshop (April 2026)**

Children's Hospital of Philadelphia, Philadelphia, PA, USA.

- Developed and taught the first hands-on one-week workshop for microbial genomics and bioinformatics for researchers at University of Ghana. The workshop session was well attended with 26 participants from multiple university departments and national and West African regional research labs. The syllabus and material are available online (<https://github.com/moustafa-lab-org/Microbial-Genomics-Journey-Workshop-Ghana26>).

### **Microbial Genomes and Metagenomes Workshop (December 2025)**

PennCHOP Microbiome Symposium, Philadelphia, PA, USA.

- Developed and taught a hands-on one-day workshop for Strain Tracking in the Microbiome for researchers at UPenn and CHOP. The workshop session was well attended with 30 participants from multiple university and hospital departments. The syllabus and material are available online (<https://github.com/kylebittinger/strain-tracking-in-the-microbiome>).

### **Microbial Genomes and Metagenomes Workshop (December 2024)**

PennCHOP Microbiome Symposium, Philadelphia, PA, USA.

- Developed and taught a hands-on one-day workshop for microbial genomes and metagenomes for researchers at UPenn and CHOP. The workshop session was well attended with 40 participants from multiple university and hospital departments. The syllabus and material are available online (<https://github.com/kylebittinger/microbial-genomes-and-metagenomes>).

### **Microbial Genomics Journey Workshop (March 2024)**

Children's Hospital of Philadelphia, Philadelphia, PA, USA.

- Developed and taught the first hands-on one-day workshop for microbial genomics and bioinformatics for researchers at University of Botswana. The workshop session was well attended with 60 participants from multiple university departments and national research labs. The syllabus and material are available online (<https://github.com/ahmedmagds/Microbial-Genomics-Journey-Workshop-Gabs24>).

### **Microbial Genomics Journey Workshop (Sep-Dec 2023)**

Children's Hospital of Philadelphia, Philadelphia, PA, USA.

- Developed and taught the first hands-on workshop for microbial genomics and bioinformatics for researchers at Penn and CHOP. For three months, the workshop sessions were well attended with an average of 20 participants every week. The syllabus and material are available online (<https://github.com/ahmedmagds/Microbial-Genomics-Journey-Workshop-Sep-2023>).

### **Microbial Genomics Journey Workshop (Feb-May 2023)**

Children's Hospital of Philadelphia, Philadelphia, PA, USA.

- Developed and taught the first hands-on workshop for microbial genomics and bioinformatics for researchers at Penn and CHOP. For three months, the workshop sessions were well attended with an average of 20 participants every week. The syllabus and material are available online (<https://github.com/ahmedmagds/Microbial-Genomics-Journey-Workshop-2023>).

### **College and University Teaching Training (Jan-Apr 2017)**

University of Pennsylvania, PA, USA.

- Delivered by professor of higher education and director of Minority Serving Institution (MSI) "Marybeth Gasman".
- The training focused on planning for a university course in the sciences; creating syllabus and class plan; developing a theory, philosophy of university teaching and evaluation instruments to assess teaching and learning in the classroom.
- The training also discussed diversity and individual learning styles and its relationship to learning.

### **Graduate Student Instructor of molecular biology (March-April 2012)**

Dr. Panjwani Center for Molecular Medicine and Drug Research (PCMD), Karachi University, Pakistan.

- Running lab sessions for master students about basic molecular microbiology techniques such as DNA extraction, PCR, gel electrophoresis and DNA sequencing.

### **Teaching Assistant (Sept 2009-Dec 2010)**

Department of Medicinal Chemistry, Faculty of Pharmacy, Alexandria University, Egypt.

- Taught undergraduate laboratory courses of medicinal chemistry for around 2000 students divided in classes of 100 each.
- I was preparing weekly lab sessions and managing the group discussions.
- Grading the exams and supervising the students in group projects.

## **PUBLICATIONS**

1. Strysko J, Hu W, Mochankana K, John-Thubuka J, Zankere T, Gopolang B, Theiller E, Jones SM, Tembo CV, Ntereke TD, Gatonye T, Lechiile K, Keatholetswe T, Bianco C, Coffin SE, McGann C, Bittinger K, Lautenbach E, Mannathoko N, Mokomane M, Mosepele M, Richard-

Greenblatt M, Nakstad B, Goldfarb DM, Planet P\*, **Moustafa AM\***. Integrated epidemiologic investigation and genomic confirmation of a *Klebsiella pneumoniae* neonatal sepsis outbreak in Botswana. medRxiv 2025.11.06.25339637; doi: <https://doi.org/10.1101/2025.11.06.25339637>. Accepted at PLOS Global Public Health (<http://journals.plos.org/globalpublichealth/article?id=10.1371/journal.pgph.0006468>). (\*= co-corresponding).

2. Theiller E, Rajagopol S, Higgins S, Torres DI, Napper T, Galis BE, Dash A, Qian E, Hamlette L, She Q, Tanes C, L'Etoile N, Feder A, Slotfeldt Viana A, Assis Côrtes Esteves MC, Abt MC, Coffin SE, Hayes E, Potter RF, Zackular JP, Srinivasan L, Sá Figueiredo AM\*, Planet PJ\*, **Moustafa AM\***. Empowering global disease surveillance with CURED: a tool for rapid identification of unique genomic biomarkers. mSystems 11:e01063-25. <https://doi.org/10.1128/msystems.01063-25>. (\*= co-corresponding).
3. She Q, Srinivasan L, Theiller E, Galis BE, Napper T, Feder A, Arvanitis A, Jones SM, Hayes E, Puopolo KM, Baldassano RN, David MZ, Coffin SE, Gibbs KA, Potter RF, Smith K, Harris RM, Zackular JP\*, **Moustafa AM\***, Planet PJ\*. Rapid dissemination of *Staphylococcus aureus* in the neonatal intensive care unit is associated with invasive infection. 2026. Nat Commun 17, 2550 (2026). <https://doi.org/10.1038/s41467-026-69074-z>. (\*= co-corresponding).
4. Odih EE, Abdulahi J, Amulele A, Bates M, Heinz E, Hu W, Jain K, Magobo R, Olwagan CP, Tembo J, Sonda T, Stryko J, Tigoi CC, Bittinger K, Cornick J, Foster-Nyarko E, Gumbi W, Jones SM, Musyani CL, McGann CM, **Moustafa AM**, Musicha P, Mwansa JCL, Ndumba ML, Stanton TD, Omuoyo DO, Pearse O, Phillips LT, Planet PJ, Rodrigues CMC, Secka F, Sands K, Theiller E, Zuza AM, Basu S, Chan GJ, Iregbu KC, Mazarati JB, Alemayehu SS, Walsh TR, Zahra R, Dramowski A, Fwoloshi S, Appiah-Korang L, Madrid L, Obeng-Nkrumah N, Ojok D, Wadugu BD, Whitelaw AC, Bhargava A, Jindal A, Agarwal RK, Aiken AM, Berkley JA, Coffin SE, Feasey NA, Govender NP, Hamer DH, Madhi SA, Sankar MJ, Wyres KL, Holt KE. Contribution of nosocomial transmission to *Klebsiella pneumoniae* neonatal sepsis in Africa and South Asia: an observational study of infection clusters inferred from pathogen genomics and temporal data. Accepted at PLoS Med.
5. Mmolai S, Gatonye T, Gopolang B, Tembo CV, Keatholetswe T, Coffin SE, Richard-Greenblatt M, Annavajhala MK, Hoar C, Bédard É, **Moustafa AM**, Planet PJ, Stryko J. Stewarding the hospital sink drain: a narrative review of practical approaches for controlling Gram-negative pathogens in low- and middle-income countries. 2026. Antimicrob Resist Infect Control 15, 25 (2026). <https://doi.org/10.1186/s13756-026-01701-2>.
6. Stanton TD, Keegan SP, Abdulahi JA, Amulele AV, Bates M, Heinz E, Hooda Y, Hu W, Jain K, Kanwar S, Magobo R, Olwagan CP, Tembo JM, Sonda T, Stryko J, Tigoi CC, Amin SAA, Bittinger K, Cornick J, Foster-Nyarko E, Gumbi W, Hotwani A, Iqbal N, Jones SM, Kabir F, Khan W, Musyani CL, McGann CM, Mittal V, **Moustafa AM**, Musicha P, Mwansa JCL, Ndumba ML, Odih EE, Omuoyo DO, Pearse O, Phillips LT, Planet PJ, Rasool AA, Rodrigues CMC, Sands K, Tanmoy AM, Theiller E, Zuza AM, Basu S, Chan GJ, Iregbu KC, Mazarati JB, Alemayehu SS, Walsh TR, Zahra R, Dramowski A, Fwoloshi S, Labi AK, Madrid L, Obeng-Nkrumah N, Ojok D, Wadugu BD, Whitelaw AC, Bethou A, Bhargava A, Jindal A, Nanavati RN, Prasad PS, Sastry A, Farooqi JQ, Ghanchi N, Jehan F, Khan E, Agarwal RK, Aiken AM, Berkley JA, Coffin SE, Feasey NA, Govender NP, Hamer DH, Madhi SA, Nisar MI, Saha SK, Saha S, Sankar MJ, Wyres KL, Holt KE. Distribution of capsule and O types in *Klebsiella pneumoniae* causing neonatal sepsis in Africa and South Asia: a meta-analysis of genome-predicted serotype prevalence to inform potential vaccine coverage. 2026. PLoS Med. 23(1):e1004879.

7. Kelliher JM, Mirzayi C, Bordenstein SR, Oliver A, Kellogg CA, Hatcher EL, Berg M, Baldrian P, Aljumaah M, Miller CML, Mungall C, Novak V, Palucki A, Smith E, Tabassum N, Bonito G, Brister JR, Chain PSG, Chen M, Degregori S, Dundore-Arias JP, Emerson JB, Fernandes VMC, Flores R, Gonzalez A, Hansen ZA, Jackson SA, **Moustafa AM**, Northen TR, Pariente N, Pett-Ridge J, Record S, Reji L, Reysenbach A-L, Rich VI, Richardson L, Roux S, Schriml LM, Shabman RS, Sierra MA, Sullivan MB, Sundaramurthy P, Thibault KM, Thompson LR, Tighe S, Vereen E, Eloë-Fadrosh EA. STREAMS guidelines: standards for technical reporting in environmental and host-associated microbiome studies. 2025. *Nat Microbiol.* 1–10.
8. Stryko J, Thela T, Feder A, Thubuka J, Machiya T, Mkubwa J, Mochankana K, Tiroyakgosi C, Kgomanyane K, Ntereke TD, Zankere T, Lechiile K, Gatonye T, Tembo CV, Vurayai M, Mannathoko N, Mokomane M, **Moustafa AM**, Goldfarb DM, Richard-Greenblatt M, McGann C, Coffin SE, Nakstad B, Cancedda C, Lautenbach E, Bogoshi D, Smith AM, Planet PJ. Carbapenem-resistant *Acinetobacter baumannii* at a hospital in Botswana: detecting a protracted outbreak using whole genome sequencing. 2025. *Microbiol Spectr.* 14(1):e01768-25.
9. McGann CM, Stryko J, Bayani O, Moorad B, Ntlhako N, Nwako N, Hu W, Tanes C, **Moustafa AM**, Jones SM, Nakstad B, Gezmu AM, Arscott-Mills T, Goldfarb DM, Steenhoff AP, Richard-Greenblatt M, Machiya T, Thela T, Mokomane M, Paganotti GM, Vurayai M, Zalot M, Boustany M, Planet PJ, Coffin S, Bittinger K. Effect of maternal HIV status on the early neonatal microbiome. 2025. *J Pediatr Infect Dis Soc.* 14(10):piaf076.
10. L'Etoile N, Sundararaman SA, Theiller E, Miller JJ, **Moustafa AM**, Steenhoff AP, Odom John AR. Pocapavir for neonatal enteroviral sepsis: limitations of rapid viral sequencing on treatment decisions. 2025. *ASM Case Rep.* 1(6):e00008-25.
11. Kelliher JM, Aljumaah M, Bordenstein SR, Brister JR, Chain PSG, Dundore-Arias JP, Emerson JB, Fernandes VMC, Flores R, Gonzalez A, Hansen ZA, Hatcher EL, Jackson SA, Kellogg CA, Madupu R, Miller CML, Mirzayi C, **Moustafa AM**, Mungall C, Oliver A, Pariente N, Pett-Ridge J, Record S, Reji L, Reysenbach A-L, Rich VI, Richardson L, Schriml LM, Shabman RS, Sierra MA, Sullivan MB, Sundaramurthy P, Thibault KM, Thompson LR, Tighe S, Vereen E, Eloë-Fadrosh EA. Microbiome data management in action workshop: Atlanta, GA, USA, June 12–13, 2024. 2025. *Environ Microbiome.* 20(1):40.
12. Free M, Soper N, Slaughter JC, Feder A, Bianco C, **Moustafa AM**, Planet PJ, Creech CB, Thomsen I. Evolution of strain diversity and virulence factor repertoire in pediatric *Staphylococcus aureus* isolates. 2025. *PLoS One.* 20(7):e0326353.
13. Lemons JMS, Firman J, Mahalak KK, Liu L, Narrowe AB, Higgins S, **Moustafa AM**, Baudot A, Deyaert S, Van den Abbeele P. The effect of sodium benzoate on the gut microbiome across age groups. 2025. *Foods.* 14(17):2949.
14. Liu L, Lemons JMS, Firman J, Mahalak KK, Chetty VJ, Narrowe AB, Higgins S, **Moustafa AM**, Baudot A, Deyaert S, Van den Abbeele P. Hydrogen sulfide has a minor impact on human gut microbiota across age groups. 2025. *Sci.* 7(3):102.
15. Rotsaert C, Minnebo Y, Duysburgh C, Liu L, Mahalak KK, Firman J, Mattei LM, **Moustafa AM**, Bittinger K, Hu W, Marzorati M, Michiels J, Van de Wiele T. Digestive parameters and gut microbiota load and composition along the in vivo piglet gastrointestinal tract. 2025. *J Anim Sci.* 103:skaf107. <https://doi.org/10.1093/jas/skaf107>
16. Lemons JMS, Narrowe AB, Firman J, Mahalak KK, Liu L, Higgins S, **Moustafa AM**, Baudot A, Deyaert S, Van den Abbeele P. The food additive butylated hydroxyanisole minimally affects the human gut microbiome ex vivo. 2025. *Food Chem.* 473:143037. <https://doi.org/10.1016/j.foodchem.2025.143037>.



17. Martínez JRW, Alcalde-Rico M, Jara-Videla E, Reyes J, Carvajal LP, Rincon S, Ríos R, Diaz L, Quesille-Villalobos A, Riquelme-Neira R, Rivas L, **Moustafa AM**, Hanson B, Undurraga EA, Olivares-Pacheco J, García P, Araos R, Planet PJ, Arias CA, Munita JM. Geographic divergence of methicillin-resistant *Staphylococcus aureus* ST5-SCCmecI in the aftermath of a major earthquake and tsunami: impact of a plasmid harboring heavy metal resistance genes. 2025. mBio0:e03669-24. <https://doi.org/10.1128/mbio.03669-24>
18. Johanna M.S. Lemons, Adrienne B. Narrowe, Jenni Firman, Karley K. Mahalak, LinShu Liu, Higgins S, **Moustafa AM**, Aurélien Baudot, Stef Deyaert, Pieter Van den Abbeele. The food additive butylated hydroxyanisole minimally affects the human gut microbiome ex vivo, Food Chemistry, Volume 473, 2025,143037,ISSN 0308-8146, <https://doi.org/10.1016/j.foodchem.2025.143037>
19. Zirbes CF, Feder A, Pamatmat AJ, Bartels AR, Pitcher NJ, Rozen AL, Teresi M, Krogh J, Regan M, Arnold EA, Hill JJ, Reinhardt LD, Oberto CL, Boyken L, Reeb VC, **Moustafa AM**, Planet PJ, Fischer AJ. Genetic Concordance of *Staphylococcus aureus* From Oropharyngeal and Sputum Cultures in People with Cystic Fibrosis. *Pediatr Pulmonol*. 2025 Jan; 60(1). <https://doi.org/10.1002/ppul.27475>
20. Mulcrone PL, Narrowe A, Jones S, Hu W, Bittinger K, **Moustafa AM**, Firman, J. (2025). Monitoring the impact of chlorinated drinking water on the developing gut microbiota of adolescent mice using genetic profiling. *International Journal of Environmental Studies*, 1–18. <https://doi.org/10.1080/00207233.2024.2444847>
21. Liu S, Rodriguez J, Munteanu V, Ronkowski C, Sharma N, Alser M, Andreace F, Blekhman R, Błaszczyk D, Chikhi R, Crandall K, Della Libera K, Francis D, Frolova A, Gancz A, Huntley N, Jaiswal P, Kosciolk T, Łabaj P, Łabaj W, Luan T, Mason C, **Moustafa AM**, Muralidharan H, Mutlu O, Ghiasi N, Rahnavard A, Sun F, Tian S, Tierney B, Van Syoc E, Vicedomini R, Zackular J, Zelikovsky A, Zielińska K, Ganda E, Davenport E (R), Pop M, Koslicki D, Mangul S. Analysis of metagenomic data. *Nat Rev Methods Primers* 5, 5 (2025). <https://doi.org/10.1038/s43586-024-00376-6>
22. Yamana TK, Rajagopal S, Hall DC Jr., **Moustafa AM**, Feder A, Ahmed A, Bianco C, Harris R, Coffin S, Campbell AE, Pei S, Mell JC, Planet PJ, Shaman J. A two-variant model of SARS-COV-2 transmission: Estimating the characteristics of a newly emerging strain. *BMC Infect Dis*. 2024; 24 (938). <https://doi.org/10.1186/s12879-024-09823-x>
23. Alameh MG, Semon A, Bayard NU, Pan YG, Knox J, Dwivedi G, Glover RC, Rangel PC, Tanes C, Bittinger K, She Q, Maslanka JR, Planet PJ, **Moustafa AM**, Davis B, Chevrier A, Beattie M, Ni H, Blizard G, Furth EE, Mach RH, Lavertu M, Sellmyer M, Tam Y, Abt MC, Weissman D, Zackular JP. A multivalent mRNA-LNP vaccine protects against *Clostridioides difficile* infection. *Science* 2024. DOI: 10.1126/science.adn4955
24. Urso A, Monk I, Predella C, Fok Lung TW, Theiller E, Boylan J, Perelman S, Baskota S, **Moustafa AM**, Lewis I, Torres V, Chen YT, Lohia GK, Howden B, Stinear T, Dorrello NV, Prince A. *Staphylococcus aureus* adapts to exploit collagen-derived proline during chronic infection. *Nat Microbiol* 2024. <https://doi.org/10.1038/s41564-024-01769-9>
25. Mahalak KK, Liu L, Bobokalonov JT, Narrowe AB, Firman J, Bittinger K, Hu W, Jones SM, **Moustafa AM**. Supplementation with soluble or insoluble rice-bran fibers increases short-chain fatty acid producing bacteria in the gut microbiota in vitro. *Frontiers in Nutrition*. 2024; 11:16125. <https://doi.org/10.3389/fnut.2024.1304045>
26. Liu L, Mahalak KK, Bobokalonov JT, Narrowe AB, Firman J, Lemons JMS, Bittinger K, Hu W, Jones SM, **Moustafa AM**. Impact of Ivermectin on the Gut Microbial Ecosystem. *International Journal of Molecular Sciences*. 2023; 24(22):16125. <https://doi.org/10.3390/ijms242216125>



27. Smith AB, Specker JT, Hewlett KK, Scoggins TR, Knight M, Lustig AM, Li Y, Evans KM, Guo Y, She Q, Christopher MW, Garrett TJ, **Moustafa AM**, Van Tyne D, Prentice BM, Zackular JP. Liberation of host heme by Clostridioides difficile-mediated damage enhances Enterococcus faecalis fitness during infection. mBio. 2023, <https://doi.org/10.1128/mbio.01656-23>.
28. Free M, Soper N, Slaughter JC, Feder A, Bianco C, **Moustafa AM**, Planet P, Creech CB, Thomsen I. Evolution of Invasive and Colonizing Pediatric *Staphylococcus aureus* Isolates, *Journal of the Pediatric Infectious Diseases Society*, Volume 12, Issue Supplement\_1, November 2023, Pages S3–S4, <https://doi.org/10.1093/jpids/piad070.007>.
29. Esteves MAC, Viana AS, Viçosa GN, Botelho AMN, **Moustafa AM**, Mansoldo FRP, Ferreira ALP, Vermelho AB, Ferreira-Carvalho BT, Planet PJ, Figueiredo AMS. RdJ detection tests to identify a unique MRSA clone of ST105-SCCmecII lineage and its variants disseminated in the metropolitan region of Rio de Janeiro. *Front. Microbiol.* 2023, 14:1275918, <https://doi.org/10.3389/fmicb.2023.1275918>
30. Pitcher NJ, Feder A, Bolden N, Zirbes CF, Pammatmat AJ, Boyken L, Hill JJ, Bartels AR, Thurman AL, Reeb VC, Porterfield HS, **Moustafa AM**, Planet PJ, Fischer AJ. Parallel evolution of linezolid-resistant *Staphylococcus aureus* in patients with cystic fibrosis. *Microbiol. Spectr.* 2023, 11(5), <https://doi.org/10.1128/spectrum.02084-23>
31. Martínez JRW, Planet PJ, Spencer-Sandino M, Rivas L, Díaz L, **Moustafa AM**, Quesille-Villalobos A, Riquelme-Neira R, Alcalde-Rico M, Hanson B, Carvajal LP, Rincón S, Reyes J, Lam M, Calderon J, Araos R, García P, Arias CA, Munita JM. Dynamics of the MRSA Population in A Chilean Hospital: A Phylogenomic Analysis (2000-2016). *Microbiol. Spectr.* 2023, <https://doi.org/10.1128/spectrum.05351-22>
32. Bianco C, **Moustafa AM**, O'Brien K, Martin M, Read TD, Kreiswirth B, Planet PJ. Pre-epidemic evolution of USA300 clade and molecular key for classification. *Front in Cell and Infect Micro.* 2023, 13, <https://doi.org/10.3389/fcimb.2023.1081070>
33. Firman J, Liu L, Mahalak KK, Hu W, Bittinger K, **Moustafa AM**, Narrowe A, Tomasula P. An in-vitro analysis of how lactose modifies the gut microbiota structure and function of adults in a donor-independent manner. *Front in Nutr.* 2023 Jan 26. ; 9:1040744. eCollection 2022. doi: 10.3389/fnut.2022.1040744.
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35. Viana AS, Botelho AM, Feder A, **Moustafa AM**, Silva DNS, Martini CL, Ferreira ALP, Silva-Carvalho<sup>1</sup> MC, Carvalho BT, Planet PJ, Figueiredo AMS. High frequency of triclosan low-dose resistance among CC5 MRSA and risk of misclassification of the SCCmec into types. *J. Antimicrob. Chemother.* 2022, dkac322. <https://doi.org/10.1093/jac/dkac322>.
36. Marques AD, Sherrill-Mix S, Everett J, Reddy S, Hokama P, Roche AM, Hwang Y, Glascock A, Whiteside SA, Graham-Wooten J, Khatib LA, Fitzgerald AS, **Moustafa AM**, Bianco C, Rajagopal S, Helton J, Deming R, Denu L, Azad A, Kitt E, Coffin SE, Newbern C, Mell JC, Planet PJ, Badjatia N, Richards B, Wang Z, Cannuscio CC, Strelau K, Jaskowiak-Barr A, Cressman L, Loughrey S, Ganguly A, Feldman MD, Collman RG, Rodino KG, Kelly BJ, Bushman FD. SARS-CoV-2 Variants Associated with Vaccine Breakthrough in the Delaware Valley through Summer 2021. *mBio.* 2022, 13(1). doi: 10.1128/mbio.03788-21.
37. Saenkham P, Hyre AN, Hanson BS, Donati GL, Adams LG, Ryan C, Londoño A, **Moustafa AM**, Planet PJ, Subashchandrabose S. Copper resistance promotes fitness of Methicillin-Resistant *S. aureus* during urinary tract infection. *mBio.* 2021, 12(5). doi: 10.1128/mBio.02038-21.

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39. Viana AS\*, Botelho AM\*, **Moustafa AM\***, Boge CLK, Ferreira ALP, Silva-Carvalho MC, Guimarães MA, Costa BSS, de Mattos MC, Maciel SP, Echevarria-Lima J, Narechania A, O'Brien K, Ryan C, Gerber JS, Ferreira-Carvalho BT, Figueiredo AMS, Planet PJ. Multidrug-resistant MRSA associated with bacteremia and monocyte evasion, Rio de Janeiro, Brazil. *Emerg Infect Dis.* 2021, 27(11): 21-0097. (\*=1<sup>st</sup> coauthorship)
40. **Moustafa AM**, Planet PJ. Jumping a moving train: SARS-CoV-2 evolution in real time. *J Pediatr Infect Dis.* 2021, piab051.
41. Miller JJ, Shah IT, Hatten J, Barekatin Y, Mueller EA, **Moustafa AM**, Edwards RL, Dowd CS, Planet PJ, Muller FL, Jez JM, Odom John AR. Structure-guided microbial targeting of antistaphylococcal prodrugs. *eLife.* 2021, 10:e66657.
42. **Moustafa AM**, Bianco C, Denu L, Ahmed A, Coffin SE, Neide B, Everett J, Reddy S, Rabut E, Deseignora J, Feldman MD, Rodino KG, Bushman F, Harris RM, Chang Mell J, Planet PJ. Comparative Analysis of Emerging B.1.1.7+E484K SARS-CoV-2 Isolates. *Open Forum Infect Dis.* 2021, 8(7):ofab300.
43. Perelman SS, James DBA, Boguslawski KM, Nelson CW, Ilmain JK, Zwack EE, Prescott RA, Mohamed A, Tam K, Chan R, Narechania A, Pawline MB, Vozhilla N, **Moustafa AM**, Kim SY, Dittmann M, Ekiert DC, Bhabha G, Shopsin B, Planet PJ, Koralov SB, Torres VJ. Genetic variation of staphylococcal LukAB toxin determines receptor tropism. *Nat Microbiol.* 2021, (6):731-745.
44. **Moustafa AM**, Velusamy SK, Denu L, Narechania A, Fine DH, Planet PJ. Adaptation by Ancient Horizontal Acquisition of Butyrate Metabolism Genes in *Aggregatibacter actinomycetemcomitans*. *mBio.* 2021, 12(2). doi: 10.1128/mBio.03581-20.
45. **Moustafa AM**, Lal A and Planet PJ. Comparative genomics in infectious disease. *Curr Opin Microbiol* 2020, 53:61-70.
46. **Moustafa AM** and Planet PJ. WhatsGNU: a tool for identifying proteomic novelty. *Genome Biol* 2020, 21:58.
47. Riquelme S\*, Lozano C\*, **Moustafa AM\***, Liimatta K, Tomlinson KL, Britto C, Khanal S, Gill SK, Narechania A, Azcona-Gutiérrez JM, DiMango E, Saénz Y, Planet PJ, Prince A. CFTR-PTEN-dependent mitochondrial metabolic dysfunction promotes *Pseudomonas aeruginosa* airway infection. *Sci Transl Med* 2019, 11(499):eaav4634. (\*=1<sup>st</sup> coauthorship)
48. Acker KP, Wong Fok Lung T, West E, Craft J, Narechania A, Smith H, O'Brien K, **Moustafa AM**, Lauren C, Planet PJ, Prince A. Strains of *Staphylococcus aureus* that colonize and infect skin harbor mutations in metabolic genes. *iScience* 2019, 19:281-290.
49. Botelho AM, Cerqueira e Costa MO, **Moustafa AM**, Beltrame CO, Ferreira FA, Côrtes MF, Costa BSS, Silva DNS, Bandeira PT, Lima NCB, Souza RC, de Almeida LGP, Vasconcelos ATR, Narechania A, Ryan C, O'Brien K, Kolokotronis SO, Planet PJ, Nicolás MF, Figueiredo AMS. Local diversification of methicillin-resistant *Staphylococcus aureus* ST239 in South America after its rapid worldwide dissemination. *Front Microbiol* 2019, 10:82.
50. Zapotoczna M, Pelicioli-Riboldi G, **Moustafa AM**, Dickson E, Narechania A, Morrissey JA, Planet PJ, Holden MTG, Waldron KJ, Geoghegan JA. Mobile genetic element-encoded hypertolerance to copper protects *Staphylococcus aureus* from killing by host phagocytes. *mBio* 2018, 9(5):e00550-18.
51. **Moustafa AM** and Bennett MD. Development of loop-mediated isothermal amplification-based diagnostic assays for detection of *Pasteurella multocida* and hemorrhagic septicemia-associated *P. multocida* serotype B:2. *AJVR* 2017, 78(2):134-43.

52. **Moustafa AM**, Ali SN, Bennett MD, Hyndman TH, Robertson ID, Edwards J. A case-control study of haemorrhagic septicaemia in buffaloes and cattle in Karachi, Pakistan, in 2012. *Transbound Emerg Dis* 2015. doi: 10.1111/tbed.12393.
53. **Moustafa AM**, Seemann T, Gladman S, Adler B, Harper M, Boyce JD, Bennett MD. Comparative genomic analysis of Asian haemorrhagic septicaemia-associated strains of *Pasteurella multocida* identifies more than 90 haemorrhagic septicaemia-specific genes. *PLoS ONE* 2015, 10(7):e0130296.
54. **Moustafa AM**, Bennett MD, Edwards J, Azim K, Mesaik MA, Choudhary MI, Pathanasophon P, Worarach A, Ali Q, Abubakar M, Anjum R. Molecular typing of haemorrhagic septicaemia-associated *Pasteurella multocida* isolates from Pakistan and Thailand using multilocus sequence typing and pulsed-field gel electrophoresis. *Res Vet Sci* 2013, 95:986-90.

## GRANTS

- 1U54AG089323-01, NIH, Collman, Ron. Role: Co-lead of the Biospecimen Analysis Core (02/01/2025-01/31/2030). The Oro-Respiratory-Gut Virome Axis over Space and Time. Total Award Amount (including indirects): \$20,213,896.
- 1U54AI198184-01, NIH, Fisher, Brian. Role: Co-Investigator (09/25/2025-07/31/2030). Pediatric Fungal Network Study of Rare Invasive Fungal DisEases in Immunocompromised Pediatric Patients. Total Award Amount (including indirects): \$8,586,135.
- 1R01AI185544-01A1, NIH, Srinivasan, Lakshmi. Role: Co-Investigator (07/17/2025-07/16/2030) Whole Genome Tracking of Pathogen Transmission and Infection in the Neonatal ICU (NICU). Total Award Amount (including indirects): \$4,293,693.
- 1DP2AI192737-01, NIH, Annavajhala, Medini. Role: Co-Investigator (07/01/2025-06/30/2030) Decoding multidrug-resistant pathogen dynamics for clinically-relevant wastewater surveillance. Total Award Amount (including indirects): \$2,684,758.
- R01DE033033-01A1, NIH, Teles, Flavia. Role: Co-Investigator (05/17/2024-04/30/2029) Defining Dysbiosis and Mechanisms of Periodontitis Progression and Stability. Total Award Amount (including indirects): \$1,256,724.
- 1U19AI174998-01, NIH, Bushman, Rick. Role: Co-Investigator (02/01/2023-01/31/2028) MVACS--mRNA Vaccines for Clostridioides difficile Suppression. Total Award Amount (including indirects): \$7,649,770.
- CHOP Research Institute OMICS Initiative Award, Moustafa, Ahmed. Role: Primary Investigator (01/01/2024-6/30/2025) Empowering Global Disease Surveillance with the CURED Tool for Rapid Identification of Unique Clonal Biomarkers. Total Award Amount (including indirects): \$150,000.
- CHOP Research Institute OMG Award, Planet, Paul. Role: Co-Investigator (07/01/2023-06/30/2025) High resolution dynamics of *S. aureus* gastrointestinal colonization, transmission, and infection using metagenomics, geospatial-omics and culturomics. Total Award Amount (including indirects): \$100,000.
- Gates Foundation, Coffin, Susan. Role: Co-Investigator (01/01/2024-12/31/2024) Sequencing Neonatal Isolates of *Klebsiella* from Botswana. Total Award Amount (including indirects): \$98,766.
- NIH, Moustafa, Ahmed. Role: Primary Investigator (08/17/2023-08/17/2024) Sequencing multiple microbiome samples. Total Award Amount (including indirects): \$96,865.

- NIH, Moustafa, Ahmed. Role: Primary Investigator (01/22/2024-08/22/2024) Sequencing multiple microbiome samples. Total Award Amount (including indirects): \$19,010.
- PennCHOP Microbiome Program Pilot Grant, Moustafa, Ahmed. Role: Primary Investigator (07/01/2023-06/30/2024) A Microbiome Cartography (Micro Cart) for the CHOP NIICU. Total Award Amount: \$40,000.
- GRT-00002526-07, CHOP Research Institute Next Generation Sequencing Award, Moustafa, Ahmed. Role: Primary Investigator (01/01/2023-12/31/2023) A strain level pathogen tracking pipeline for personalized patient care and infection control. Total Award Amount (including indirects): \$100,000.
- The Foerderer Foundation at the Children's Hospital of Philadelphia. McGann, Carolyn. Role: Other (7/1/2022-06/30/2023). Characterizing Maternal Influences on the Evolving Microbiome and Colonization Dynamics of Neonates. Total Award Amount (including indirects): \$49,932.
- USDA Microbiome Award 12305B22P0095, USDA Moustafa, Ahmed. Role: Primary Investigator (6/20/2022 - 6/19/2023). Microbiome Metagenomic Shotgun Sequencing and Whole Genome Sequencing of *K. pneumoniae*. Total Award Amount (including indirects): \$ 116,688.
- CDC200-2021-11102, CDC Planet, Paul. Role: Co-Investigator (05/17/21-05/16/23). Monitoring Diversity in SARS-CoV-2 Genomes for Tracking Emerging Variants, Measuring Impact of Mitigation Strategies, and Gauging Clinical Outcomes in Pediatric Patients. Total Award Amount (including indirects): \$1,279,777.

## FEATURED WORK

1. The Science Behind the Spread: Researchers Aim to Reduce Bacterial Burden in NICUs. CHOP Cornerstone Blog, January 2026. <https://www.research.chop.edu/cornerstone-blog/the-science-behind-the-spread-researchers-aim-to-reduce-bacterial-burden-in-nicus>.
2. Understanding Klebsiella pneumoniae strains to protect newborns from sepsis. LSHTM news, January 2026. <https://www.lshtm.ac.uk/research/centres/amr/news/477526/understanding-klebsiella-pneumoniae-strains-protect-newborns-sepsis>.
3. CHOP-Penn Team Is Unraveling the Human Virome Mystery. CHOP Cornerstone Blog, November 2025. [https://www.research.chop.edu/cornerstone-blog/chop-penn-team-is-unraveling-the-human-virome-mystery?utm\\_source=linkedin&utm\\_medium=organic](https://www.research.chop.edu/cornerstone-blog/chop-penn-team-is-unraveling-the-human-virome-mystery?utm_source=linkedin&utm_medium=organic).
4. Transformative Gifts Accelerate Microbiome Research at CHOP. CHOP Cornerstone Blog, September 2025. <https://www.research.chop.edu/cornerstone-blog/transformative-gifts-accelerate-microbiome-research-at-chop>.
5. Center for Microbial Medicine Leverages 'New Frontier' for Pediatric Patients. CHOP Cornerstone Blog, August 2024. <https://www.research.chop.edu/cornerstone-blog/center-for-microbial-medicine-leverages-new-frontier-for-pediatric-patients>.
6. Arcus Enables COVID-19 Variant Research: Q&A with Scott Haag, PhD. CHOP Cornerstone Blog, June 2022. <https://www.research.chop.edu/cornerstone-blog/arcus-enables-covid-19-variant-research-qa-with-scott-haag-phd>.
7. Multidrug-Resistant MRSA, Rio de Janeiro. CDC Emerging Infectious Diseases podcast, November 2021. <https://tools.cdc.gov/medialibrary/index.aspx#/media/id/438267>.
8. Moustafa AM et al., 2021. Comparative Analysis of Emerging B.1.1.7+E484K SARS-CoV-2 Isolates. Open Forum Infect Dis. 8(7):ofab300. Discussed in August 2021 in The Atlantic (The Coronavirus Could Get Worse: <https://www.theatlantic.com/science/archive/2021/08/coronavirus-evolution-vaccines/619875/>).

9. From Mutations to Mask Mandates, New Insights Into SARS COV-2 Diversity. CHOP Cornerstone Blog, May 2021. <https://www.research.chop.edu/cornerstone-blog/from-mutations-to-mask-mandates-new-insights-into-sars-cov-2-diversity>.
10. Novel Tool Helps Track the Evolution of SARS-CoV-2: Q&A with the Planet Lab. CHOP Cornerstone Blog, Sep 2020. <https://www.research.chop.edu/cornerstone-blog/novel-tool-helps-track-the-evolution-of-sars-cov-2-qa-with-the-planet-lab>.
11. Bard *et al.*, 2021. Emergence of SARS-CoV-2 variants of concern in the pediatric population of the United States. <https://doi.org/10.1101/2021.05.22.21257660>. Discussed in Reuters, The Washington Post and Forbes among others.

## SOFTWARES

1. She Q, Joeseeph Zackular, Moustafa AM, Planet PJ (2025) Thresher, Children's Hospital of Philadelphia, Philadelphia PA (<https://github.com/microbialARC/THRESHER>).
2. She Q, Joeseeph Zackular, Moustafa AM, Planet PJ (2025) DAMAGE, Children's Hospital of Philadelphia, Philadelphia PA (<https://github.com/microbialARC/DAMAGE>).
3. Theiller E, Qian E, Feder A, She Q, Viana AS, Sá Figueiredo AM, Planet PJ, Moustafa AM (2024) CURED, Children's Hospital of Philadelphia, Philadelphia PA (<https://github.com/microbialARC/CURED>).
4. Moustafa AM, Theiller E, Feder A, Narechania A, Planet PJ (2024) Redcarpet, Children's Hospital of Philadelphia, Philadelphia PA (<https://github.com/microbialARC/Redcarpet>).
5. **Moustafa AM** & Planet PJ (2020) GNUVID, Children's Hospital of Philadelphia, Philadelphia PA (<https://github.com/ahmedmagds/GNUVID>).
6. **Moustafa AM** & Planet PJ (2019) WhatsGNU, Children's Hospital of Philadelphia, Philadelphia PA (<https://github.com/ahmedmagds/WhatsGNU>).

## SKILLS

### Informatics and laboratory

- Laboratory techniques (classical microbiological techniques and molecular biology techniques such as DNA extraction, PCR, Quantitative PCR, NGS, nanopore sequencing, loop mediated isothermal amplification (LAMP), biofilm assays, microscopy and cloning).
- Bioinformatics tools for microbial genomics (Some selected tools: BEAST, RAxML, ClonalFrameML, SPAdes, ABySS, Velvet, Prokka, RAST, Mash, MUMmer, Burrows-Wheeler Aligner, Mauve, BRIG, Snippy, Artemis, PHAST, GraPhlAn, SplitsTree, Samtools, SnpEff, Roary, Scoary, Mesquite, SNPGenie, ABRicate, iTOL, Qiime2).
- Extensive experience with UNIX command line, high-performance computing, cloud computing, and algorithmic implementation using existing software libraries and de novo scripting in Python.
- Implementing machine learning approaches in genomics.
- Good working experience with Bioconda, Biopython, R, scikit-learn, pandas and SQL.
- Univariable and multivariable epidemiological analyses (moderate).

## Languages

- Arabic, English, and moderate written and spoken command of German.

## PROFESSIONAL LEADERSHIP, SERVICE & HONORS

- Program Committee Member, The ASM Bioinformatics, Genomics and Big Data Conference (ASM BIG)—formerly ASM NGS (2026-2029). <https://asm.org/events/asm-big/scientific-program/program-committee>.
- PennCHOP Microbiome Annual Symposium Planning Committee (2023-present).
- CHOP 2023 Scientific Symposium Planning Committee (2023).
- Co-chair of Communication Committee and member of the leadership for the Microbiome Centers Consortium (March 2022-present) <https://microbiomecenters.org/leadership/>.
- Award from the National Human Genome Research Institute to defray the cost of programming course at CSHL (\$1,000) (2018).
- Travel award from Murdoch University to attend the International Pasteurellaceae Conference, Prato, Italy (2014).
- Murdoch University PhD scholarship covering tuition fees and stipend (2011-2015).
- Travel and Training Award in a Hungarian community pharmacy from the International Pharmaceutical Student's Federation (2008).

## INVITED TALKS

1. Enabling Precision Surveillance: A Genomics-Guided Approach to Rapidly Detect Circulating Pathogenic Lineages. Beyond Antibiotics: Emerging Strategies for Combating Bacterial Infection (May 2026), Breckenridge, CO, United States.
2. From Local Strain Collections to AllTheBacteria: Linking Local Isolates with Global Genomic Data for Pneumococcal Investigation. Ghana Pneumococcal Conference (April 2026), University of Ghana, Accra, Ghana.
3. Linking Hospital Outbreaks to Global Trends: Microbial Genomics and Big Data for Precision Pathogen Surveillance. Grand Rounds (November 2025), Korle Bu Teaching Hospital, Accra, Ghana.
4. From Local NICUs to Global Public Health: Leveraging Microbial Genomics and Big Data for Precision Surveillance. UCI Microbiome Center (June 2025), University of California, Irvine, Irvine, CA.
5. Innovative Tools and Strategies for Outbreak Preparedness: Integrating Environmental Microbiomes and Prospective Pathogen Collection in Hospital Settings", 1st Workshop on Computational Advances in Metagenomics Research and Applications (CAMErA 2025) in conjunction with the 13th International Conference on Computational Advances in Bio and Medical Sciences (January 2025), Georgia State University, Atlanta, GA
6. Empowering Global Disease Surveillance: Leveraging Microbial Genomics, Big Data Mining, and Environmental Microbiome Analysis to Understand Infectious Diseases. **Qatar MENA Microbiome Conference (October 2024)**, Sovereign Fund M42, Dubai, United Arab Emirates.
7. Empowering Global Disease Surveillance: Leveraging Microbial Genomics, Big Data Mining, and Environmental Microbiome Analysis to Understand Infectious Diseases. **Qatar MENA Microbiome Conference (October 2024)**, Department of Veterinary, United Arab Emirates University, Al Ain, United Arab Emirates.

8. Empowering Global Disease Surveillance: Leveraging Microbial Genomics, Big Data Mining, and Environmental Microbiome Analysis to Understand Infectious Diseases. **Qatar MENA Microbiome Conference (October 2024)**, Alqassimi Hospital, Emirates Health Services, Sharjah, United Arab Emirates.
9. Empowering Global Disease Surveillance: Leveraging Microbial Genomics, Big Data Mining, and Environmental Microbiome Analysis to Understand Infectious Diseases. **Qatar MENA Microbiome Conference (September 2024)**, Doha, Qatar.
10. Redcarpet: A Tool for Rapid Recombination Detection in *Staphylococcus aureus* and Other Species Amidst Expanding Genomic Databases. 19th International Symposium on Staphylococci and Staphylococcal Infections (ISSSI) (August 2024), Perth, Australia.
11. Invited to Microbiome Data Management Workshop at ASM Microbe 2024 organized by the National Microbiome Data Collaborative (June 2024), Atlanta, GA
12. Building a Comprehensive Strain Library and Unraveling the Mysteries of Infectious Diseases Through Innovative Bioinformatics and Genomic Big Data. Grand Rounds, Sir Ketumile Masire Teaching Hospital (March 2024), Gaborone, Botswana.
13. Building a Comprehensive Strain Library and Unraveling the Mysteries of Infectious Diseases Through Innovative Bioinformatics and Genomic Big Data. National Health Laboratory (March 2024), Gaborone, Botswana.
14. Empowering Global Disease Surveillance: Leveraging Microbial Genomics and Big Data Mining to Understand Infectious Diseases. Botswana-Harvard Partnership (March 2024), Gaborone, Botswana.
15. Building a Comprehensive Strain Library and Unraveling the Mysteries of Infectious Diseases Through Innovative Bioinformatics and Genomic Big Data. Dean's Lecture Series, School of Medicine, University of Botswana (March 2024), Gaborone, Botswana.
16. Building a Comprehensive Strain Library and Unraveling the Mysteries of Infectious Diseases Through Innovative Bioinformatics and Genomic Big Data. Department of Pediatrics, The Chris Hani Baragwanath Hospital (March 2024), Johannesburg, South Africa.
17. Empowering global disease surveillance—The CURED tool for rapid identification of unique clonal biomarkers. Genome Informatics Meeting (December 2023), Cold Spring Harbor Laboratory, NY, USA.
18. Building a Comprehensive Strain Library and Unraveling the Mysteries of Infectious Diseases Through Innovative Bioinformatics and Genomic Big Data. CHOP DBHi (November 2023), Philadelphia, PA, USA.
19. Redcarpet: A Rapid Alignment-Free Database-Driven Recombination Detection Tool for Bacterial Genomes. CHOP Research Symposium (May 2023), Philadelphia, PA, USA.
20. *Microbial Genomics Big-Data Mining and its Applications in Understanding Infectious Diseases*. Lecturer. Drexel Microbiology and Immunology Department Seminar (January 2023), Philadelphia, PA, USA.
21. *Alignment-free Recombination Detection Using Genomic Database Distributions of Exact Protein Matches*. ASM Conference on Rapid Applied Microbial Next-Generation Sequencing and Bioinformatic Pipeline (October 2022), Baltimore, MD, USA.
22. *Exact match compression for identification of genomic novelty in the Staphylococci*. Staphylococcal Diseases, Gordon Research Conference (August 2019), Barcelona, Spain.
23. *Biofilm formation of A. aphrophilus in bone and joint infections: role of the tad locus*. MedVetPathogens (October 2018), Prato, Italy.
24. *A PhD story*. The Annual Scientific Symposium (August 2015), Alexandria, Egypt.



25. *Can you stop drinking milk?* The Three Minute Thesis Competition (August 2014), Perth, Western Australia, **Third Place**.
26. *Comparative genomics analysis of Asian HS-associated strains of Pasteurella multocida*. The International Pasteurellaceae Conference (May 2014), Prato, Italy.
27. *Multilocus sequencing typing*. The Application of DNA Sequencing in Medicine and Biotechnology Workshop (February 2012), Karachi, Pakistan.

## ABSTRACTS/POSTERS

1. Higgins S, Wilson N, Toner A, Yadav A, Marques AD, Joffe S, Bittinger K, Bushman FD, Collman RG, Moustafa AM. Defining the Limit of Detection for Viral Pathogens in Plasma Using DNA and RNA Metagenomic Sequencing: Implications for Ethical Reporting and Return of Results. NIH [Human Virome Program](#) Annual Meeting, Poster in March 2026.
2. Theiller E, Galis BE, Jones S, Napper T, She Q, Srinivasan L, Zackular JP, Planet PJ, Moustafa AM: Unraveling the Epidemiological Dynamics of Staphylococcus epidermidis in Pediatric Bacteremia. GRS Staphylococcal Diseases Conference, Barcelona, Spain, Talk in August 2025.
3. Moustafa AM\*, Theiller E, Lal A, Yamali S, Feder A, Narechania A, Planet PJ: Redcarpet: A Tool for Rapid Recombination Detection in Staphylococcus aureus and Other Species Amidst Expanding Genomic Databases. GRS Staphylococcal Diseases Conference, Barcelona, Spain, Poster in August 2025.
4. Elgarten CW, Wilson N, Thornton C, Blumenstock J, Morrison K, Stein M, Moustafa AM, Bittinger K, Fisher BT: Oral microbiome features correlate with mucositis in pediatric patients receiving chemotherapy. PennCHOP Microbiome Symposium, Philadelphia, PA, Poster in December 2024.
5. Stryko J, Zankere T, Hu W, Gopolang B, Tembo CV, Ntereke TD, Gatonye T, Mogotsi N, Keatholetswe T, Kgomanyane K, John-Thubuka J, Mochankana K, Nakstad B, Coffin S, McGann C, Bittinger K, Lautenbach E, Mannathoko N, Richard-Greenblatt M, Goldfarb D, Moustafa A: Using comparative genomics to detect clusters of circulating Klebsiella pneumoniae among hospitalized neonates: Botswana, 2022 - 2024. Klebsiella Epidemiology and Biology Symposium 2024, Paris, France, Podium November 2024.
6. She Q, Feder A, Galis BE, Jones SM, Gibbs K, Hayes E, Smathers S, Thompson K, Harris R, Zackular JP, Moustafa AM, Srinivasan L, Planet PJ: Phylogenomic analysis of colonizing and invasive Staphylococcus aureus in a neonatal intensive care unit reveals high levels of transmission of invasive strains. ASM NGS Conference, Washington DC, Podium October 2024.
7. McGann C, Stryko J, Zankere T, Hu W, Gopolang B, Tembo CV, Ntereke TD, Gatonye T, Mogotsi N, Keatholetswe T, Kgomanyane K, John-Thubuka J, Mochankana K, Nakstad B, Coffin S, Steenhoff A, Bittinger K, Planet PJ, Lautenbach E, Mannathoko N, Richard-Greenblatt M, Goldfarb D, Moustafa A: Genomic epidemiology of Klebsiella pneumoniae in a hospital in Southern Africa, 2021-2024. CHOP Global Health Symposium, Philadelphia, PA, Poster October 2024.
8. Rotsaert C, Minnebo Y, Duysburgh C, Mahalak K, Firman J, Liu LS, Moustafa AM, Mattei L, Bittinger K, Marzorati M, Michiels J, Van de Wiele T. Structure and function of the gut microbiota of the in vivo piglet gastrointestinal tract. American Chemical Society (August 2024), Colorado, USA.

9. **Moustafa AM**, Lal A, Theiller E, Feder A, Narechania A, Planet PJ. Redcarpet: A Tool for Rapid Recombination Detection Amidst Expanding Genomic Databases. Genome Informatics Meeting (December 2023), Cold Spring Harbor Laboratory, NY, USA.
10. Theiller E, Galis B, Higgins S, Jones S, She Q, Planet PJ, Zackular JP, **Moustafa AM**. MicrobialARC: Genomics Driven Initiative to Improve Personalized Patient Care and Pathogen Surveillance. CHOP Research Symposium (May 2023), Philadelphia, PA, USA.
11. She Q, Feder A, Galis B, Jones S, Gibbs K, Hayes E, Smathers S, Thompson K, Harris R, Zackular JP, **Moustafa AM**, Srinivasan L, Planet PJ. Phylogenomic analysis of colonizing and invasive *Staphylococcus aureus* in a neonatal intensive care unit reveals high levels of transmission of invasive strain. CHOP Research Symposium (May 2023), Philadelphia, PA, USA. **Best Poster Prize**.
12. Rajagopal S, Feder A, Martinez K, Arvanitis AG, Bianco C, Denu L, Helton J, Smith C, Newbern C, Schell J, Ahmed A, Chang-Mell J, **Moustafa AM**, Planet PJ. Trends in molecular diversity of SARS-CoV-2 in children and adults in Philadelphia throughout the pandemic. CHOP 2022 poster day and scientific symposium (May 2022), Philadelphia, PA, USA. **Best Poster Prize**.
13. Firman J, Mulcrone P, Hu W, Bittinger K, Liu L, Xiao W, **Moustafa AM**. The effect of chlorinated drinking water on the gut microbiota: an in vivo analysis. American Chemical Society (ACS) Fall meeting (August 2022), Chicago, IL, USA.
14. **Moustafa AM**, Lal A, Feder A, Narechania A, Planet PJ. Alignment-free Recombination Detection Using Genomic Database Distributions of Exact Protein Matches. 6<sup>th</sup> Annual MidAtlantic Bioinformatics Conference (November 2021), Philadelphia, PA, USA.
15. Pitcher N, Zirbes C, **Moustafa AM**, Planet PJ, Thurman A, Pezzulo A, Porterfield H, Reeb V, Fischer AJ. Linezolid-resistant *Staphylococcus aureus* in patients with cystic fibrosis. North American Cystic Fibrosis Conference (NACFC) (November 2021), San Antonio, TX, USA.
16. **Moustafa AM**, Alexander S, Pitcher N, Goldberg J, Fischer AJ, Planet PJ. Longitudinal Evolution and Adaptation of *Staphylococcus aureus* in Cystic Fibrosis. North American Cystic Fibrosis Conference (NACFC) (November 2021), San Antonio, TX, USA.
17. Souza S, Faria A, **Moustafa AM**, Planet P, Merquior V, Teixeira L. Populational structure and evolutionary scenario of the multidrug resistant pathogen *Enterococcus faecium* in hospitals of Rio de Janeiro, Brazil. 31<sup>st</sup> European Congress of Clinical Microbiology and Infectious Diseases (ECCMID) (July 2021).
18. **Moustafa AM**, Planet PJ. *SARS-CoV-2 Diversity in Six US States Using GNUVID Tool*. ASM Conference on Rapid Applied microbial Next-Generation Sequencing and Bioinformatic Pipelines 2020 Virtual Conference (December 2020), USA.
19. **Moustafa AM**, Planet PJ. *Rapid whole genome sequence typing reveals multiple waves of SARS-CoV-2 spread*. IDWeek 2020 Virtual Conference (October 2020), USA.
20. **Moustafa AM**, Planet PJ. *Rapid whole genome sequence typing reveals multiple waves of SARS-CoV-2 spread*. Genome Informatics Meeting (September 2020), Wellcome Genome Campus, Hinxton, UK.
21. **Moustafa AM**, Planet PJ. *Rapid whole genome sequence typing reveals multiple waves of SARS-CoV-2 spread*. COVID/SARS-CoV-2 Rapid Research Reports Virtual Meeting (August 2020), Cold Spring Harbor Laboratory, NY, USA.
22. Bouso JM, Ehrlich R, **Moustafa AM**, Graf E, Narechania A, Mell JC, Planet PJ. *Chronic Nontuberculous Mycobacterial Infections in Cystic Fibrosis Are Characterized by Recombination and Polyclonality*. American Thoracic Society (2020), Virtual; C106. Basic and translational discoveries in TB and mycobacterial diseases.

23. Tomlinson K, Dach F, Riquelme SA, Wong T, **Moustafa AM**, Planet PJ, Kahl BC, Prince A. *In vivo adaptation of S. aureus to the lung is driven by itaconate, a mitochondrial immune metabolite*. American Thoracic Society (2020), Virtual; D102. Bacterial infections: immunity and basic mechanisms.
24. **Moustafa AM**, Planet PJ. *WhatsGNU: a tool for identifying proteomic novelty*. Genome Informatics Meeting (November 2019), Cold Spring Harbor Laboratory, NY, USA.
25. **Moustafa AM**, Riquelme SA, Lozano C, Narechania A, Saenz Y, Prince A, Planet PJ. *Molecular clock analysis of longitudinal isolates of Pseudomonas aeruginosa reveals time-ordered adaptations to the host*. North American Cystic Fibrosis Conference (NACFC) (November 2019), Nashville, TN, USA.
26. Bouso JM, Ehrlich R, **Moustafa AM**, Graf E, Narechania A, Mell JC, Planet PJ. Evolutionary phylogenomics and proteomics of Nontuberculous Mycobacteria. North American Cystic Fibrosis Conference (NACFC) (November 2019), Nashville, TN, USA.
27. Dach F, Riquelme SA, Tomlinson K, Liimatta K, **Moustafa AM**, Planet PJ, Kahl BC, Prince A. *In vivo evolution of CF Staphylococcus aureus strains in response to immune cell signaling metabolites*. North American Cystic Fibrosis Conference (NACFC) (November 2019), Nashville, TN, USA.
28. Martínez JRW, Diaz L, Rojas M, Rios R, Hanson B, Rivas LM, Spencer M, **Moustafa AM**, Bralic RA, Peters A, Reyes J, Carvajal LP, Luna C, Salles M, Alvarez C, Labarca J, Seas C, Guzmán M, Planet PJ, Arias CA, Munita J. 556. *Phylogenomic Epidemiology of Methicillin-Resistant Staphylococcus aureus (MRSA) Chilean-Cordobes Clone in Latin America*. ID Week (October 2019), Open Forum Infectious Diseases, Volume 6(S2), Washington DC, USA.
29. **Moustafa AM**, Planet PJ. *WhatsGNU: a tool for identifying proteomic novelty*. Symposium on Advances in Genomics, Epidemiology, and Statistics (SAGES) (June 2019), University of Pennsylvania, Philadelphia, PA, USA. **Best Poster Prize**.
30. Riquelme SA, Lozano C, **Moustafa AM**, Narechania A, DiMango E, Saenz Y, Planet PJ, Prince AS. *Bacterial Metabolic Adaptation Causes Chronic Lung Infection in Cystic Fibrosis*. American Thoracic Society (May 2019), Dallas, TX; D21. Host response to lung infections.
31. Bouso JM, **Moustafa AM**, Ehrlich R, Narechania A, Mell JC, Planet PJ. Evolutionary Phylogenomics Identified Hypermutable Isolates of Nontuberculous Mycobacteria in A Cystic Fibrosis Chronic Infection. American Thoracic Society (May 2019), Dallas, TX; D34 Cystic Fibrosis and bronchiectasis: Clinical and Mechanistic Studies/Thematic.
32. **Moustafa AM**, Planet PJ, Hyndman TH, Harper M, Edwards J, Boyce JD, Christensen H. *The first draft genomes of E:2 (African) serotype of Pasteurella multocida associated with haemorrhagic septicaemia disease*. MedVetPathogens (October 2018), Prato, Italy.
33. Riquelme S, Lozano C, **Moustafa AM**, Gill SK, Narechania A, DiMango E, Saenz Y, Planet PJ, Prince A. *Succinate drives Pseudomonas aeruginosa adaptation to the cystic fibrosis airway*. North American Cystic Fibrosis Conference (NACFC) (October 2018), Denver, CO, USA.
34. Bouso JM, Ehrlich R, **Moustafa AM**, Ryan C, Graf E, Mell JC, Planet PJ. *Host adaptation of NTM in a chronically-infected CF patient*. North American Cystic Fibrosis Conference (NACFC) (October 2018), Denver, CO, USA.
35. **Moustafa AM**, Sheth CT, Russo M, Narechania A, Mott C, Dreyer A, Greene S, Marek L, DeSalle R, Swami S, Figurski D, Planet PJ. *Tight adherence by A. aphrophilus & osteoarticular infections*. ASM Microbe (June 2018), Atlanta, GA, USA.
36. **Moustafa AM**, Bennett MD. *Development of rapid LAMP diagnostic test specific for haemorrhagic septicaemia disease*. The International Pasteurellaceae Conference (May 2014), Prato, Italy. **Best Poster Prize**.

37. **Moustafa AM**, Ali SN, Bennett MD, Hyndman TH, Robertson ID, Edwards J. *A retrospective case-control study of haemorrhagic septicaemia disease in Karachi, Pakistan in 2012*. The International Pasteurellaceae Conference (May 2014), Prato, Italy.
38. **Moustafa AM**, Bennett MD, Edwards J, Azim K, Mesaik MA, Choudhary MI, Pathanasophon P, Worarach A, Ali Q, Abubakar M, Anjum R. *Molecular typing of haemorrhagic septicaemia-associated Pasteurella multocida isolates from Pakistan and Thailand using multilocus sequence typing and pulsed-field gel electrophoresis*. The Murdoch University Poster Day (November 2013), Perth, Western Australia. **Excellence in Biomedical Sciences Poster Prize**.

## WORKSHOPS & COURSES

1. Participated in the AMR Genomics Hackathon (<https://github.com/AMR-genomics-hackathon-2026>) at Cambridge University hosted by the ESCMID Study Group on Epidemiological Markers-AMR working group (March 2026).
2. Invited to Microbiome Data Management Workshop at ASM Microbe 2024 organized by the National Microbiome Data Collaborative, Atlanta, GA, 2024.
3. Penn-Wharton Commercialization workshop (May 4-6, 2023). The Wharton School of the University of Pennsylvania Mack Institute for Innovation Management.
4. Culturally Aware Mentoring (March 10, 2023). Jordan Medical Education Center (JMEC) University of Pennsylvania, PA. Training provided by PSOM ITMAT.
5. LEEP: Leadership Essentials for Early-Career PIs (October 2022-March 2023). Training provided by: The CHOP Research Institute.
6. CAMB 700- Topics in Microbiology (March 14, 2023). Visiting faculty.
7. PSOM: Writing Letters of Recommendation for Trainees (Dec 19, 2022).
8. PSOM: Clarifying Performance Expectations (Dec 14, 2022).
9. Research Mentor Training: Effective Communication and Aligning Expectations (November 2022). Jordan Medical Education Center (JMEC), University of Pennsylvania, PA. Training provided by PSOM ITMAT.
10. DeepLearning.AI TensorFlow Developer Specialization by deeplearning.ai on Coursera. Certificate earned at 01/10/2021.
11. Sequences, Time Series and Prediction by deeplearning.ai on Coursera. Certificate earned at 01/10/2021.
12. Natural Language Processing in TensorFlow by deeplearning.ai on Coursera. Certificate earned at 01/05/2021.
13. Convolutional Neural Networks in TensorFlow by deeplearning.ai on Coursera. Certificate earned at 12/13/2020.
14. Machine Learning by Stanford University on Coursera, Certificate earned 10/04/2020.
15. TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning by deeplearning.ai on Coursera. Certificate earned 04/26/2020.
16. Weeklong metagenomics virtual training course, Chris Mason's Lab at Cornell Medical Center (March 2020), Trelify Virtual-Internship.
17. Pangenome Metadata Scopeathon, San Diego State University (January 2020), San Diego, CA, USA.
18. *Taming the BEAST (Bayesian analysis of molecular sequences) five-day workshop*, Quest University (August 12-16<sup>th</sup>, 2019), Squamish, BC, Canada.

19. *Hands-on Microfluidics workshop for fabrication of soft lithography devices*, Singh Center for Nanotechnology (March, 2019), University of Pennsylvania, PA, USA.
20. *Programming for Biology* two-week intensive course, Cold Spring Harbor Laboratory (October 14-30<sup>th</sup>, 2018), NYC, NY, USA.
21. One Million Arab Coders program - Data Analyst track (August 2018), Udacity.
22. Essentials of Staphylococcal Genetics Workshop, University of Nebraska Medical Center (July 2018), Omaha, Nebraska, USA.
23. *Metagenomics Analysis Workshop*, PennCHOP Microbiome Program (November 2017), University of Pennsylvania, PA, USA.
39. *Winter School in Mathematical and Computational Biology*, Australian Research Council (ARC) Centre of Excellence in Bioinformatics (July 2012), University of Queensland, Brisbane, Australia.
40. *International Workshop on Bioinformatics: Database Mining and High Throughput Genomic Analyses*, Sindh Institute of Urology and Transplantation (SIUT) (March 2012), Karachi, Pakistan.
41. *Application of DNA Sequencing in Medicine and Biotechnology Workshop*, International Center for Chemical and Biological Sciences (February 2012), Karachi University, Pakistan.

## REVIEWER SERVICES

- Invited reviewer by Dr Françoise Thibaud-Nissen, lead of the NCBI Prokaryotic Annotation and Curation Resources team, to test and give feedback on the new improvements done to the annotation of genes on archaeal and bacterial genomes (June 2021) at NCBI, NIH, USA.
- Invited reviewer by Dr Nuala O'Leary to test the new NCBI datasets command line tool from Nuala O'Leary group (February-March 2020) at NCBI, NIH, USA.
- Serving as Area Chair for the 16th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB 2025), held in Philadelphia, where I oversaw the quality of peer reviews, coordinated with reviewers, recruited additional experts when needed, and authored meta-reviews to provide balanced recommendations and ensure a rigorous and inclusive evaluation process (August 2025).
- Invited reviewer for the following journals; eLife, Frontiers in Microbiology, Frontiers Veterinary Science, Veterinary Microbiology, Cladistics, mSphere, Gene, PLOS ONE, The Veterinary Journal, Antimicrobial Resistance and Infection Control, Letters in Applied Microbiology, Nature Scientific Reports, Nature Communications Biology, Computational and Structural Biotechnology Journal, MDPI Viruses, MDPI Pathogens.

## PUBLIC OUTREACH AND ADVOCACY

- *American Society of Microbiology Meeting with offices of Sen. Cory Booker (D-NJ), Rep. Frank Pallone (D-NJ), Sen. Bob Menendez (D-NJ), and Rep. Dwight Evans (D-PA)*. December 2022, Washington DC, USA.

## REFERENCES

- **Dr Robert N. Baldassano** (CMM Co-Director and Current Mentor)

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